

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409983
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Guan; Weixian et al.

Mobile cooling box with hinge module

Abstract

A mobile cooling box having a box main body and at least one lid for opening the box and providing access to the inside of the box. The at least one lid is pivotally attached to the box main body of at least two hinge modules. Each hinge module comprises a pin module having a hinge pin with a front end, a rear end, a longitudinal axis about which the lid is pivotable and predominantly a smooth outer surface of a cylindrical shape. Each hinge module further comprises a bearing module having a hinge bearing accommodating the hinge pin. The hinge pin laterally extends with its front end into the hinge bearing. Thus, during pivoting the lid with respect to the box main body an axis of the hinge bearing remains co-linear with the longitudinal axis of the hinge pin.

Inventors: Guan; Weixian (JinWan, CN), Wang; Peng (JinWan, CN), Tan; Wenmin (JinWan, CN), Chen; Yuelong (JinWan, CN)

Applicant: Dometic Sweden AB (Solna, SE)

Family ID: 71104593

Assignee: Dometic Sweden AB (Solna, SE)

Appl. No.: 18/502801

Filed: November 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240067414 A1	Feb. 29, 2024

Foreign Application Priority Data

DE	102019200070.3	Jan. 04, 2019
----	----------------	---------------

Related U.S. Application Data

Publication Classification

Int. Cl.: **B65D43/16** (20060101); **B65D25/28** (20060101); **B65D43/22** (20060101); **F25D3/08** (20060101)

U.S. Cl.:

CPC **B65D43/165** (20130101); **B65D25/2841** (20130101); **B65D43/22** (20130101); **F25D3/08** (20130101);

Field of Classification Search

CPC: B65D (43/165); B65D (43/164); B65D (43/163); B65D (43/169); B65D (43/161); B65D (43/16); B65D (43/22); B65D (25/2841); B65D (25/2838); B65D (25/2835); B65D (25/28); A45C (11/20); F25D (3/08); F25D (3/06); F25D (3/02); Y10S (220/9151)

USPC: 220/592.03; 220/592.02; 220/915.2; 220/915.1; 220/254.3; 220/254.2; 220/254.5; 220/844; 220/843; 220/848; 220/836; 220/810; 220/830; 220/827; 220/826; 62/457.7; 62/457.2; 62/457.1

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1691468	12/1927	Albert	N/A	N/A
2012800	12/1934	Allen	N/A	N/A
2492601	12/1948	Steel	N/A	N/A
2642728	12/1952	Thomaras	N/A	N/A
2715817	12/1954	Brodheim	N/A	N/A
2766901	12/1955	Sunko	N/A	N/A
2926504	12/1959	Hellinger	N/A	N/A
3035733	12/1961	Knapp	N/A	N/A
3338068	12/1966	Piker	N/A	N/A
3796063	12/1973	Wulke et al.	N/A	N/A
3938694	12/1975	Archuleta	N/A	N/A
3979007	12/1975	Thornbloom, Jr.	N/A	N/A
4162029	12/1978	Esch et al.	N/A	N/A
D255312	12/1979	Uyeda	N/A	N/A
D256587	12/1979	Tatsumi	N/A	N/A
4328676	12/1981	Reed	N/A	N/A
4351165	12/1981	Gottsegen et al.	N/A	N/A
4468932	12/1983	Bullard	N/A	N/A
4757920	12/1987	Harootian, Jr. et al.	N/A	N/A
4802344	12/1988	Livingston et al.	N/A	N/A
4866945	12/1988	Bender et al.	N/A	N/A

4886176	12/1988	Steakley	N/A	N/A
5024471	12/1990	Kahl et al.	N/A	N/A
5301508	12/1993	Kahl et al.	N/A	N/A
5319937	12/1993	Fritsch et al.	N/A	N/A
D349007	12/1993	Costello	N/A	N/A
5407218	12/1994	Jackson	N/A	N/A
5676296	12/1996	Masters	N/A	N/A
5860281	12/1998	Coffee et al.	N/A	N/A
5897435	12/1998	Thomas et al.	N/A	N/A
6027249	12/1999	Bielinski	N/A	N/A
6308518	12/2000	Hunter	N/A	N/A
6349845	12/2001	Duncan	N/A	N/A
6354101	12/2001	Levitin	N/A	N/A
6354104	12/2001	Feagin	N/A	N/A
6474097	12/2001	Treppedi et al.	N/A	N/A
D497517	12/2003	Reuter	N/A	N/A
D500057	12/2003	Reuter	N/A	N/A
6964447	12/2004	McNamee	N/A	N/A
D516099	12/2005	Maruyama	N/A	N/A
7143601	12/2005	Jimenez	N/A	N/A
D534771	12/2006	Zorn	N/A	N/A
7178673	12/2006	Miller	N/A	N/A
7415794	12/2007	Thompson	N/A	N/A
7617699	12/2008	Parmely	N/A	N/A
7913849	12/2010	Pedrazzi et al.	N/A	N/A
D640102	12/2010	Hawkins	N/A	N/A
8210819	12/2011	Wiebe et al.	N/A	N/A
8312816	12/2011	Vaccarella	N/A	N/A
D712720	12/2013	Seiders	N/A	N/A
D712721	12/2013	Seiders	N/A	N/A
D712722	12/2013	Seiders	N/A	N/A
D712723	12/2013	Seiders	N/A	N/A
D714125	12/2013	Seiders	N/A	N/A
8827109	12/2013	Sheehan	N/A	N/A
8910819	12/2013	Seiders	N/A	N/A
D722474	12/2014	Seiders	N/A	N/A
D722475	12/2014	Seiders	N/A	N/A
D732348	12/2014	Seiders et al.	N/A	N/A
D732349	12/2014	Seiders et al.	N/A	N/A
D732350	12/2014	Seiders et al.	N/A	N/A
D732899	12/2014	Seiders et al.	N/A	N/A
9137232	12/2014	Eschbatch et al.	N/A	N/A
9139352	12/2014	Seiders et al.	N/A	N/A
9163871	12/2014	Costello	N/A	N/A
9187232	12/2014	Seiders	N/A	N/A
9220250	12/2014	Davis, Sr.	N/A	N/A
9232290	12/2015	Besay	N/A	N/A
D752347	12/2015	Seiders et al.	N/A	N/A
9282797	12/2015	Soto	N/A	N/A
9320938	12/2015	Belmore	N/A	N/A

D792486	12/2016	Li	N/A	N/A
D802028	12/2016	Li	N/A	N/A
D802029	12/2016	Li	N/A	N/A
D802630	12/2016	Li	N/A	N/A
9951986	12/2017	Turner	N/A	N/A
9982894	12/2017	Philip et al.	N/A	N/A
D820049	12/2017	Ahlstrom	N/A	N/A
D836993	12/2018	Meda	N/A	N/A
D836994	12/2018	Meda	N/A	N/A
D844386	12/2018	Ahlstrom	N/A	N/A
D887788	12/2019	Meda	N/A	N/A
D888503	12/2019	Meda	N/A	N/A
10717499	12/2019	Street	N/A	N/A
D894043	12/2019	Meda	N/A	N/A
D901986	12/2019	Meda	N/A	N/A
D920743	12/2020	Meda et al.	N/A	N/A
D927938	12/2020	Meda et al.	N/A	N/A
D929471	12/2020	Meda et al.	N/A	N/A
D935280	12/2020	Thelin et al.	N/A	N/A
D954764	12/2021	Meda et al.	N/A	N/A
11396406	12/2021	Guan	N/A	N/A
11415355	12/2021	Thelin et al.	N/A	N/A
2001/0039807	12/2000	Mogil	N/A	N/A
2002/0095947	12/2001	Treppedi et al.	N/A	N/A
2003/0019873	12/2002	Nam	N/A	N/A
2003/0042266	12/2002	Young et al.	N/A	N/A
2003/0106895	12/2002	Kalal	N/A	N/A
2004/0178208	12/2003	Leba et al.	N/A	N/A
2004/0195793	12/2003	Sullivan et al.	N/A	N/A
2005/0274726	12/2004	Boggs et al.	N/A	N/A
2005/0279123	12/2004	Maldonado et al.	N/A	N/A
2005/0279750	12/2004	Barquist et al.	N/A	N/A
2007/0101754	12/2006	Maldonado	N/A	N/A
2007/0125100	12/2006	Shoenfeld	N/A	N/A
2008/0011759	12/2007	Hsu	N/A	N/A
2009/0126419	12/2008	Yoon et al.	N/A	N/A
2010/0251730	12/2009	Whillock, Sr.	N/A	N/A
2011/0226785	12/2010	Sakell	N/A	N/A
2013/0068776	12/2012	Patterson	N/A	N/A
2013/0334238	12/2012	Goforth	N/A	N/A
2015/0008242	12/2014	Kpabar, Jr.	N/A	N/A
2015/0021106	12/2014	LaRosa	N/A	N/A
2015/0114024	12/2014	Grepper	N/A	N/A
2015/0241107	12/2014	Mech	N/A	N/A
2015/0322698	12/2014	Seiders	N/A	N/A
2015/0353263	12/2014	Seiders et al.	N/A	N/A
2016/0101924	12/2015	Mitchell et al.	N/A	N/A
2016/0187046	12/2015	Chen et al.	N/A	N/A
2017/0023290	12/2016	DeMuth	N/A	N/A
2017/0108253	12/2016	Monroy	N/A	N/A

2017/0254578	12/2016	Kriesel	N/A	N/A
2018/0015938	12/2017	Defrancia	N/A	N/A
2018/0100682	12/2017	Nilsen	N/A	N/A
2018/0141718	12/2017	Ahlstrom	N/A	N/A
2018/0147913	12/2017	Bergin	N/A	N/A
2018/0201432	12/2017	Harding	N/A	N/A
2018/0306479	12/2017	Vazquez	N/A	N/A
2019/0161240	12/2018	Ahlstrom	N/A	N/A
2019/0308795	12/2018	Juneau	N/A	N/A
2020/0062454	12/2019	Duong et al.	N/A	N/A
2020/0216229	12/2019	Weixian	N/A	N/A
2020/0217571	12/2019	Weixian	N/A	N/A
2020/0217574	12/2019	Thelin	N/A	N/A
2020/0217575	12/2019	Thelin	N/A	N/A
2020/0217576	12/2019	Weixian	N/A	N/A
2020/0217578	12/2019	Wang	N/A	N/A
2022/0313012	12/2021	Ganahl	N/A	B65D 25/32
2022/0355984	12/2021	Guan et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201712770	12/2016	AU	N/A
201712777	12/2016	AU	N/A
201712780	12/2016	AU	N/A
201712782	12/2016	AU	N/A
201712791	12/2016	AU	N/A
201712799	12/2016	AU	N/A
201712802	12/2016	AU	N/A
201712803	12/2016	AU	N/A
201712785	12/2016	AU	N/A
201716789	12/2016	AU	N/A
201716791	12/2016	AU	N/A
201717667	12/2017	AU	N/A
201717669	12/2017	AU	N/A
201717670	12/2017	AU	N/A
201717674	12/2017	AU	N/A
201717676	12/2017	AU	N/A
201717662	12/2017	AU	N/A
201816061	12/2018	AU	N/A
201910011	12/2018	AU	N/A
201913598	12/2018	AU	N/A
201913601	12/2018	AU	N/A
201913603	12/2018	AU	N/A
201913604	12/2018	AU	N/A
201913607	12/2018	AU	N/A
201913624	12/2018	AU	N/A
201913626	12/2018	AU	N/A
201913627	12/2018	AU	N/A

201913631	12/2018	AU	N/A
201913633	12/2018	AU	N/A
202012545	12/2019	AU	N/A
202012548	12/2019	AU	N/A
2019284128	12/2019	AU	N/A
2019284129	12/2019	AU	N/A
2019284130	12/2019	AU	N/A
2019284131	12/2019	AU	N/A
2019284133	12/2019	AU	N/A
2019284134	12/2019	AU	N/A
1888356	12/2006	CN	N/A
101074057	12/2006	CN	N/A
101208568	12/2007	CN	N/A
206488529	12/2016	CN	N/A
107804580	12/2017	CN	N/A
207081259	12/2017	CN	N/A
208312853	12/2018	CN	N/A
106233083	12/2018	CN	N/A
305240920	12/2018	CN	N/A
110191654	12/2018	CN	N/A
209893773	12/2019	CN	N/A
305553099	12/2019	CN	N/A
305751278	12/2019	CN	N/A
306079204	12/2019	CN	N/A
666767	12/1937	DE	N/A
20110247	12/2001	DE	N/A
202004010081	12/2003	DE	N/A
19981898	12/2004	DE	N/A
202007001638	12/2006	DE	N/A
202009015164	12/2009	DE	N/A
102011100722	12/2011	DE	N/A
202014000386	12/2013	DE	N/A
202013007655	12/2014	DE	N/A
102014221784	12/2015	DE	N/A
102019200065	12/2018	DE	N/A
102019200067	12/2018	DE	N/A
102019200068	12/2018	DE	N/A
102019200070	12/2018	DE	N/A
112017005901	12/2018	DE	N/A
102019200063	12/2019	DE	N/A
102019200064	12/2019	DE	N/A
005230802-001	12/2017	EA	N/A
005230802-002	12/2017	EA	N/A
001470413-0001-02	12/2018	EA	N/A
005948601-0001-30	12/2018	EA	N/A
1000577	12/1999	EP	N/A
2772704	12/2013	EP	N/A
432256	12/1934	GB	N/A
2419177	12/2005	GB	N/A
H1194417	12/1998	JP	N/A

4362188	12/2008	JP	N/A
1020050042093	12/2004	KR	N/A
9936324	12/1998	WO	N/A
2014131679	12/2013	WO	N/A
2017071654	12/2016	WO	N/A
2018095957	12/2017	WO	N/A
2018231826	12/2017	WO	N/A
2018233611	12/2017	WO	N/A
2019206219	12/2017	WO	N/A

OTHER PUBLICATIONS

US D935,281 S, 11/2021, Thelin et al. (withdrawn) cited by applicant

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/711,905 mailed on Sep. 24, 2021. cited by applicant

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Sep. 24, 2021. cited by applicant

Final Office Action Issued in U.S. Appl. No. 16/728,715 mailed on Oct. 7, 2021. cited by applicant

Non Final Office Action Issued in U.S. Appl. No. 16/728,604 mailed on Oct. 25, 2021. cited by applicant

Notice of Allowance Issued in U.S. Appl. No. 29/785,568 mailed on Nov. 16, 2021. cited by applicant

Corrected Notice of Allowance Issued in U.S. Appl. No. 29/785,568 mailed on Dec. 6, 2021. cited by applicant

United Kingdom Application No. 6157637 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157638 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157639 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157640 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157641 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157642 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157643 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157644 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157645 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157646 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157647 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157648 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157649 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

United Kingdom Application No. 6157650 titled "Mini Fridge" filed on Aug. 27, 2021. cited by applicant

applicant
United Kingdom Application No. 6157651 titled “Mini Fridge” filed on Aug. 27, 2021. cited by applicant
United Kingdom Application No. 6157652 titled “Mini Fridge” filed on Aug. 27, 2021. cited by applicant
United Kingdom Application No. 6157653 titled “Mini Fridge” filed on Aug. 27, 2021. cited by applicant
United Kingdom Application No. 6157654 titled “Mini Fridge” filed on Aug. 27, 2021. cited by applicant
United Kingdom Application No. 6157655 titled “Mini Fridge” filed on Aug. 27, 2021. cited by applicant
United Kingdom Application No. 6157656 titled “Mini Fridge” filed on Aug. 27, 2021. cited by applicant
Office Action Issued in German Patent Application No. 102019200070.3 mailed on Aug. 25, 2021. cited by applicant
Office Action Issued in U.S. Appl. No. 16/728,646 mailed on Dec. 21, 2021. cited by applicant
Notice of Allowance Issued in U.S. Appl. No. 16/728,715 mailed on Feb. 2, 2022. cited by applicant
Notice of Allowance Issued in U.S. Appl. No. 29/785,568 mailed on Feb. 8, 2022. cited by applicant
Non Final Office Action Issued in U.S. Appl. No. 16/728,687 mailed on Apr. 15, 2022. cited by applicant
Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on May 2, 2022. cited by applicant
Supplemental Notice of Allowance Issued in U.S. Appl. No. 16/728,604 mailed on May 13, 2022. cited by applicant
Corrected Notice of Allowance Issued in U.S. Appl. No. 16/728,715 mailed on May 16, 2022. cited by applicant
Restriction Requirement Issued in U.S. Appl. No. 16/800,512 mailed on May 24, 2022. cited by applicant
U.S. Appl. No. 17/825,374 titled “Mobile Cooling Box with Ice Maker” filed May 26, 2022. cited by applicant
Design U.S. Appl. No. 29/840,161 titled “Cooler Fender Frame” filed May 26, 2022. cited by applicant
Non-Final Office Action Issued in U.S. Appl. No. 16/728,646 mailed on May 26, 2022. cited by applicant
Design U.S. Appl. No. 29/803,817 titled “Fender Frame, Internal Closure, and Hinge of a Cooler” filed Aug. 16, 2021. cited by applicant
Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Feb. 7, 2022. cited by applicant
Decision to Grant Issued in German Application No. 102019200070.3 mailed on Mar. 2, 2022. cited by applicant
Corrected Notice of Allowance issued in U.S. Appl. No. 17/886,803 mailed on May 16, 2024. cited by applicant
Corrected Notice of Allowance issued in U.S. Appl. No. 18/464,742 mailed on May 15, 2024. cited by applicant
Corrected Notice of Allowance Issued in U.S. Appl. No. 29/697,071 mailed on Jun. 9, 2021. cited by applicant
Corrected Notice of Allowance Issued in U.S. Appl. No. 29/697,070 mailed on Jun. 9, 2021. cited by applicant

Notice of Allowance Issued in U.S. Appl. No. 29/711,905 mailed on Jul. 12, 2021. cited by applicant

Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Jul. 12, 2021. cited by applicant

Australian Patent Application No. 2020264299 entitled "Cover for a cooler" filed Nov. 4, 2020. cited by applicant

German Patent Application No. 102020213897.4 titled "Cover for a cooler" filed Nov. 4, 2020. cited by applicant

Amazon.com, ICECO insulated protective cover, pp. 1-2, first date available Oct. 11, 2019. (Year: 2019). cited by applicant

Non Final Office Action Issued in U.S. Appl. No. 16/728,657 mailed on Nov. 15, 2021. cited by applicant

Notice of Allowance Issued in U.S. Appl. No. 16/728,657 mailed on Mar. 24, 2022. cited by applicant

Notice of Allowance Issued in U.S. Appl. No. 16/728,657 mailed on Mar. 9, 2022. cited by applicant

Supplemental Notice of Allowance Issued in U.S. Appl. No. 16/728,657 mailed on May 2, 2022. cited by applicant

Notice of Allowance Issued in U.S. Appl. No. 16/728,657 mailed on Jun. 15, 2022. cited by applicant

Corrected Notice of Allowability Issued in U.S. Appl. No. 16/728,604 mailed on Jul. 8, 2022. cited by applicant

U.S. Appl. No. 17/886,803 titled "Mobile Cooling Box with Handle Module" filed Aug. 12, 2022. cited by applicant

Non-Final Office Action Issued in U.S. Appl. No. 16/728,623 mailed on Oct. 3, 2022. cited by applicant

Non-Final Office Action Issued in U.S. Appl. No. 16/800,512 mailed on Oct. 11, 2022. cited by applicant

Notice of Allowance Issued in U.S. Appl. No. 16/728,646 mailed on Oct. 24, 2022. cited by applicant

Non-Final Office Action issued in U.S. Appl. No. 17/825,374 mailed on Oct. 25, 2022. cited by applicant

Final Office Action Issued in U.S. Appl. No. 16/728,687 mailed on Nov. 28, 2022. cited by applicant

Decision to Grant Issued in German Application No. 102020209896.4 mailed on Jan. 10, 2023. cited by applicant

Final Office Action Issued in U.S. Appl. No. 16/800,512 mailed on Feb. 14, 2023. cited by applicant

Notice of Allowability Issued in U.S. Appl. No. 16/728,646 mailed on Feb. 15, 2023. cited by applicant

Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Mar. 7, 2023. cited by applicant

Corrected Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Mar. 16, 2023. cited by applicant

Non-Final Office Action issued in U.S. Appl. No. 16/728,687 mailed on Mar. 17, 2023. cited by applicant

Corrected Notice of Allowability issued in U.S. Appl. No. 16/728,646 mailed on Apr. 24, 2023. cited by applicant

Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Apr. 24, 2023. cited by applicant

Patent Certificate issued in DE Patent Application No. 102020209896.4 mailed on Apr. 20, 2023. cited by applicant

Final Office Action Issued in U.S. Appl. No. 16/728,623 mailed on May 19, 2023. cited by applicant

Corrected Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Jun. 1, 2023. cited by applicant

Non-Final Office Action issued in U.S. Appl. No. 17/886,803 mailed on May 23, 2023. cited by applicant

Notice of Allowability issued in U.S. Appl. No. 17/825,374 mailed on Jul. 26, 2023. cited by applicant

U.S. Appl. No. 18/464,742 titled “Mobile Cooling Box with Ice Maker” filed Sep. 11, 2023. cited by applicant

Restriction Requirement issued in U.S. Appl. No. 29/803,817 mailed on Aug. 15, 2023. cited by applicant

Corrected Notice of Allowability issued in U.S. Appl. No. 16/728,646 mailed on Aug. 31, 2023. cited by applicant

Office Action issued in CN Application No. 201911066985.7 mailed on Jul. 26, 2023. cited by applicant

Notice of Acceptance issued in AU Application No. 2019284129 mailed on Jan. 8, 2024. cited by applicant

Notice of Allowability issued in U.S. Appl. No. 29/803,817 mailed on Jan. 18, 2024. cited by applicant

Restriction Requirement for Design U.S. Appl. No. 29/697,070 mailed on Dec. 28, 2020. cited by applicant

Notice of Allowance Issued in U.S. Appl. No. 29/697,071 mailed on Dec. 23, 2020. cited by applicant

Examination report mailed in EU Design Application No. 005948601-0001/005948601-0030 on Jan. 9, 2019. cited by applicant

Examination report mailed in DE Patent Application No. 102019200070.3 on Sep. 24, 2019. cited by applicant

Examination report mailed in DE Patent Application No. 102019200064.9 on Sep. 27, 2019. cited by applicant

Examination report mailed in DE Patent Application No. 102019200067.3 on Oct. 8, 2019. cited by applicant

Examination report mailed in DE Patent Application No. 102019200063.0 on Oct. 15, 2019. cited by applicant

Notification to Grant Patent Right for CN Design Application No. 201930331393.8 Mailed on Oct. 29, 2019. cited by applicant

CN Design application No. 201930331393.8 entitled “Cooler” filed on Jun. 25, 2019. cited by applicant

CN Patent Application No. 201911066985.7 entitled “Cover for a Cooler” filed Nov. 4, 2019. cited by applicant

DE Design Application No. 102019200067.3 entitled “Mobile Cooling Box With Ice Maker” filed Jan. 4, 2019. cited by applicant

DE Patent Application No. 102019200063.0 entitled “Mobile Cooling Box with User Interface Module” filed Jan. 4, 2019. cited by applicant

DE Patent Application No. 102019200064.9 entitled “Mobile Cooling Box with Latch Handle Opening” filed Jan. 4, 2019. cited by applicant

DE Patent Application No. 102019200070.3 entitled “Mobile Cooling Box With Hinge Module” filed Jan. 4, 2019. cited by applicant

European Design Application No. 005948601 “Cooler” filed Jan. 4, 2019. cited by applicant
Design U.S. Appl. No. 29/666,264 entitled “Cooler” filed Oct. 11, 2018. cited by applicant
Design U.S. Appl. No. 29/650,910, filed Jun. 11, 2018 titled “Zipper Pull”. cited by applicant
AU Patent Application No. 202012545 entitled “Cover for Cooler” filed May 1, 2020. cited by applicant
AU Patent Application No. 202012548 entitled “Cover for Cooler” filed Apr. 30, 2020. cited by applicant
European Design Application No. 007847777-0002 “CFX3 Protective covers” filed Apr. 30, 2020. cited by applicant
Decision to Grant mailed in DE Patent Application No. 102019200064.9 on Jul. 31, 2020. cited by applicant
Decision to Grant mailed in DE Patent Application No. 102019200063.0 on Aug. 4, 2020. cited by applicant
Restriction Requirement for Design U.S. Appl. No. 29/697,069 mailed on Sep. 24, 2020. cited by applicant
AU Patent Application No. 2019284133 entitled “Mobile cooling box with ice maker” filed Dec. 30, 2019. cited by applicant
AU Patent Application No. 2019284131 entitled “Mobile cooling box with hinge moduler” filed Dec. 30, 2019. cited by applicant
AU Patent Application No. 2019284128 entitled “Mobile cooling box with user interface moduler” filed Dec. 30, 2019. cited by applicant
AU Patent Application No. 2019284134 entitled “Mobile Cooling Box with Latch Handle Opening” filed Dec. 30, 2019. cited by applicant
Notice of Allowance Issued in U.S. Appl. No. 29/697,069 mailed on Feb. 1, 2021. cited by applicant
Decision to Grant mailed in DE Patent Application No. 1020192000681 on Oct. 15, 2020. cited by applicant
Examination report mailed in DE Application No. 102019200065.7 on Sep. 30, 2019. cited by applicant
Examination report mailed in DE Application No. 102019200068.1 on Oct. 8, 2019. cited by applicant
DE Patent application 102020209896.4 entitled “Handle Module” filed on Aug. 5, 2020. cited by applicant
Decision to Grant mailed in DE Patent Application No. 102019200065.7 on Aug. 4, 2020. cited by applicant
Corrected Notice of Allowance Issued in U.S. Appl. No. 29/697,069 mailed on Feb. 26, 2021. cited by applicant
Corrected Notice of Allowance Issued in U.S. Appl. No. 16/728,715 mailed on Feb. 23, 2022. cited by applicant
Supplemental Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Mar. 10, 2022. cited by applicant
Notice of Allowance Issued in U.S. Appl. No. 29/785,568 mailed on Mar. 9, 2022. cited by applicant
Corrected Notice of Allowance Issued in U.S. Appl. No. 16/728,715 mailed on Mar. 9, 2022. cited by applicant
Notice of Allowance Issued in U.S. Appl. No. 16/728,604 mailed on Apr. 13, 2022. cited by applicant
Notice of Allowance Issued in U.S. Appl. No. 29/697,071 mailed on Apr. 12, 2021. cited by applicant
United States Patent Office, Notice of Allowance in U.S. Appl. No. 29/697,070 dated Apr. 28,

2021. cited by applicant
Corrected Notice of Allowance Issued in U.S. Appl. No. 29/697,070 mailed on May 19, 2021. cited by applicant
Design U.S. Appl. No. 29/785,568, filed May 26, 2021 titled “Cooler Fender Frame”. cited by applicant
Non Final Office Action Issued in U.S. Appl. No. 16/728,715 mailed on May 19, 2021. cited by applicant
Corrected Notice of Allowance for U.S. Appl. No. 29/697,070 dated Jun. 3, 2021. cited by applicant
Corrected Notice of Allowance for U.S. Appl. No. 29/697,071 dated Jun. 3, 2021. cited by applicant
Examination Report Issued for DE Application No. 102020209896.4 mailed on Apr. 19, 2021. cited by applicant
Corrected Notice of Allowance Issued in U.S. Appl. No. 29/711,905 mailed on Aug. 11, 2021. cited by applicant
Corrected Notice of Allowance Issued in U.S. Appl. No. 29/711,906 mailed on Aug. 11, 2021. cited by applicant
Restriction Requirement Issued in U.S. Appl. No. 16/728,623 mailed on Aug. 30, 2021. cited by applicant
Design U.S. Appl. No. 29/585,279, filed Nov. 22, 2016 titled Latch. cited by applicant
Design U.S. Appl. No. 29/605,186, filed May 24, 2017 filed titled Refrigerating Apparatus. cited by applicant
Design U.S. Appl. No. 29/650,906, filed Jun. 11, 2018 titled Soft Bag Cooler. cited by applicant
Design U.S. Appl. No. 29/670,885, filed Nov. 20, 2018 titled Cooler. cited by applicant
Design U.S. Appl. No. 29/670,888, filed Nov. 20, 2018 titled Cooler. cited by applicant
Design U.S. Appl. No. 29/697,069, filed Jul. 3, 2019 titled Cooler. cited by applicant
Design U.S. Appl. No. 29/697,070, filed Jul. 3, 2019 titled Cooler. cited by applicant
Design U.S. Appl. No. 29/697,071, filed Jul. 3, 2019 titled Cooler. cited by applicant
Design U.S. Appl. No. 29/708,155, filed Oct. 3, 2019 titled Latch. cited by applicant
Design U.S. Appl. No. 29/708,156, filed Oct. 3, 2019 titled Latch. cited by applicant
Design U.S. Appl. No. 29/711,905, filed Nov. 4, 2019 titled Cover for a Cooler. cited by applicant
Design U.S. Appl. No. 29/711,906, filed Nov. 4, 2019 titled Cover for a Cooler. cited by applicant
U.S. Appl. No. 16/728,646, filed Dec. 27, 2019 titled Mobile Cooling Box with Air Vents. cited by applicant
U.S. Appl. No. 16/728,604, filed Dec. 27, 2019 titled Mobile Cooling Box with Handle Module. cited by applicant
U.S. Appl. No. 16/728,715, filed Dec. 27, 2019 titled Mobile Cooling Box with Ice Maker. cited by applicant
U.S. Appl. No. 16/728,687, filed Dec. 27, 2019 titled Mobile Cooling Box with Latch Handle. cited by applicant
U.S. Appl. No. 16/728,623, filed Dec. 27, 2019 titled Mobile Cooling Box with User Interface Module. cited by applicant
U.S. Appl. No. 16/800,512, filed Feb. 25, 2020 titled Cover for a Cooler. cited by applicant
Design U.S. Appl. No. 29/743,180 filed Jul. 20, 2020 titled Cooler. cited by applicant
Dometic Product Catalog—Coolers; 2015. cited by applicant
Dometic Product Catalog—Coolers; 2016. cited by applicant
Non-Final Office Action issued in U.S. Appl. No. 16/728,623 mailed on Mar. 14, 2024. cited by applicant
Corrected Notice of Allowance issued in U.S. Appl. No. 29/840,161 mailed on Mar. 14, 2024. cited by applicant

Notice of Allowance issued in U.S. Appl. No. 17/886,803 mailed on Mar. 20, 2024. cited by applicant

U.S. Appl. No. 18/751,707 titled "Mobile Cooling Box With Handle Module" filed Jun. 24, 2024. cited by applicant

Office Action issued in CN Application No. 201911066985.7 mailed on Mar. 23, 2024. cited by applicant

U.S. Appl. No. 16/728,657 entitled "Mobile Cooling Box with Hinge Module" filed Dec. 27, 2019. cited by applicant

U.S. Appl. No. 17/872,514 titled "Mobile Cooling Box with Ice Maker" filed Jul. 25, 2022. cited by applicant

Ex-Parte Quayle Action issued in U.S. Appl. No. 17/872,514 mailed on May 5, 2023. cited by applicant

Notice of Allowability issued in U.S. Appl. No. 17/872,514 mailed on Jul. 21, 2023. cited by applicant

Supplemental Notice of Allowability issued in U.S. Appl. No. 17/872,514 mailed on Aug. 30, 2023. cited by applicant

Supplemental Notice of Allowability issued in U.S. Appl. No. 17/872,514 mailed on Sep. 13, 2023. cited by applicant

Final Office Action issued in U.S. Appl. No. 17/886,803 mailed on Oct. 10, 2023. cited by applicant

Corrected Notice of Allowability issued in U.S. Appl. No. 16/728,646 mailed on Oct. 4, 2023. cited by applicant

Supplemental Notice of Allowability issued in U.S. Appl. No. 17/872,514 mailed on Oct. 10, 2023. cited by applicant

Notice of Allowability issued in U.S. Appl. No. 16/728,687 mailed on Oct. 10, 2023. cited by applicant

Examination Report issued for AU Application No. 2019284129 mailed on Dec. 5, 2023. cited by applicant

Notice of Allowance issued in U.S. Appl. No. 29/803,817 mailed on Dec. 6, 2023. cited by applicant

Corrected Notice of Allowability issued in U.S. Appl. No. 16/728,646 mailed on Oct. 25, 2023. cited by applicant

AU Application No. 2024200685 titled "Mobile cooking box with handle module" filed on Feb. 5, 2024. cited by applicant

U.S. Appl. No. 18/420,358 titled "Mobile Cooling Box with Latch Handle Opening" filed Jan. 23, 2024. cited by applicant

Notice of Allowance issued in U.S. Appl. No. 29/803,817 mailed on Feb. 8, 2024. cited by applicant

Corrected Notice of Allowability issued in U.S. Appl. No. 29/840,161 mailed on Feb. 16, 2024. cited by applicant

Notice of Allowance issued in U.S. Appl. No. 18/464,742 mailed on Apr. 24, 2024. cited by applicant

Corrected Notice of Allowance issued in U.S. Appl. No. 29/840,161 mailed on May 1, 2024. cited by applicant

Corrected Notice of Allowance issued in U.S. Appl. No. 29/840,161 mailed on Apr. 18, 2024. cited by applicant

U.S. Appl. No. 29/933,756 titled "Internal Closure and Hinge of a Cooler" filed Mar. 21, 2024. cited by applicant

AU Patent Application No. 2025202568 entitled "Mobile cooling box with hinge moduler" filed Apr. 10, 2025. cited by applicant

Background/Summary

CLAIM TO PRIORITY (1) This continuation patent application claims priority to and benefit of, under 35 U.S.C. § 120, U.S. patent application Ser. No. 17/872,514, filed Jul. 25, 2022, which claims priority to U.S. patent application Ser. No. 16/728,657, now issued as U.S. Pat. No. 11,396,406, filed Dec. 27, 2019, which claims priority to German patent application DE 102019200070.3, filed Jan. 4, 2019, all of which is incorporated by reference herein.

BACKGROUND

(1) The present embodiments relate to an improved mobile cooling box with a lid that is pivotally openable by means of a hinge module to enable access to the inside of the mobile cooling box.

(2) Mobile cooling boxes are well-known from the state of the art. Usually, a mobile cooling box comprises a thermo-isolated housing for storing the goods to be stored and cooled inside and a lid for allowing access to the inside to put into or remove the goods from the mobile cooling box. It is within the nature of mobile cooling boxes that such boxes are not stationary but are movable by the user. Typically, such mobile cooling boxes are used for any kind of non-stationary use, like for example during outdoor activities, camping, yachting or the like to store and cool goods like food, drinks or even medical products, etc. for a certain period of time.

(3) Known mobile cooling boxes that are openable by pivoting open a lid usually have plastic hinges. Hereby, the plastic of the lid usually forms some kind of pins that engage with respective counterparts like holes at the box main body. In products of higher quality, the pins are sometimes made of metal. Fixation of the pins, especially the of the ones made of metal requires several additional parts like screws, etc. and, thus, requires additional effort and knowledge when assembling the mobile cooling box. Moreover, the pins have to be adjusted correctly during assembling process in order to achieve a correct orientation of the pins with regard to the receiving counterparts so that a smooth and correct function of the lid is guaranteed. Moreover, it is necessary to provide further force guiding and receiving means to enable the pins to withstand the forces occurring during movement of the probably quite heavy lid, i.e. during opening and closing of the lid by the user. This raises the assembling effort and makes assembling process time consuming and, thus, expensive. Moreover, by providing the plenty of parts the overall weight of the mobile cooling box is increased.

(4) Hence, although there are possibilities available in the prior art for a working hinge arrangement for the lid of a mobile cooling box which basically fulfill the requirements for such devices, taking the above drawbacks of known hinge arrangements of mobile cooling boxes into account there is indeed room for improvements in this regard.

(5) The information included in this Background section of the specification, including any references cited herein and any description or discussion thereof, is included for technical reference purposes only and is not to be regarded subject matter by which the scope of the invention is to be bound.

SUMMARY

(6) The present application discloses one or more of the features recited in the appended claims and/or the following features which alone or in any combination, may comprise patentable subject matter.

(7) The present embodiments, therefore, provide a mobile cooling box with a cost-efficient and

reliable and robust hinge construction for the lid of a mobile cooling box.

(8) In order to solve the posed problem, the present embodiments provide a mobile cooling box having a box main body and at least one lid for opening the box and providing access, for example, from above, to the inside of the box. The at least one lid may be pivotally attached to the box main body by at least two hinge modules. Each hinge module has a pin module. The pin module has a hinge pin with a front end, a rear end, a longitudinal axis about which the lid is pivotable and predominantly a smooth outer surface of a cylindrical shape, and a bearing module having a hinge bearing accommodating the hinge pin. The hinge pin laterally extends with its front end into the hinge bearing. Hence, during pivoting the lid with respect to the box main body an axis of the hinge bearing remains co-linear with the longitudinal axis of the hinge pin.

(9) According to the one embodiment, for the mounting of the pin module, the pin module further comprises an engaging portion extending from the rear end of the hinge pin and having a longitudinal axis being co-linear to that of the hinge pin. Thus, the pin module is mountable by the engaging portion.

(10) Thereby, the hinge pin is easily mountable without the need of other components or members like additional screws or bolts. Due to the co-linearity of the axes of the hinge pin and the engaging portion, the position of the pin is ensured.

(11) According to one embodiment of the mobile cooling box, the engaging portion may be a bolt portion, the engaging portion having a male thread. Thereby, mounting is easy and the connection is strong.

(12) According to another embodiment of the mobile cooling box, the pin module further comprises a backing plate between the hinge pin and the engaging portion. The backing plate is positioned in a plane perpendicular to the longitudinal axis of the hinge pin. The backing plate has a pin-side surface and a bolt-side surface. The backing plate has several advantages. It assists at the mounting of the pin module and provides for a stable position thereof ensuring that the pin withstands the forces occurring during movement of the lid while opening or closing the lid.

(13) According to a further embodiment of the mobile cooling box, the backing plate may have a circular shape. Since the complete pin module is screwed or bolted in, the circular shape of the backing plate provides for rotational symmetry. Advantageously, this makes it irrelevant at which rotational position the hinge module is mounted.

(14) According to another embodiment of the mobile cooling box, the pin module may be mounted to the box main body at a vertical surface thereof.

(15) According to a further embodiment of the mobile cooling box, the backing plate abuts with its bolt-side surface against said vertical surface of the box main body. This reliably hinders the pin module from inclining away from the position of the initial longitudinal axis over time during use and thus, contributes to the force resistance of the hinge module.

(16) According to another embodiment of the mobile cooling box, the vertical surface of the box main body has a pin module attachment portion. The pin module attachment portion comprises a bore in which the engaging portion is fastened. Further, the pin module attachment portion comprises a recess for accommodating the backing plate. The recess even more ensures the position of the pin module and hinders inclination of the hinge pin during extensive use. The recess preferably has the same circular shape as the backing plate and, by screwing the pin module in, the backing plate fits with and sinks into the recess.

(17) According to a further embodiment of the mobile cooling box, the recess has a depth corresponding to the thickness of the backing plate. Thus, the transition from said vertical surface of the box main body to the surface of the pin-side surface of the backing plate is essentially flush. This provides for a smooth surface and fewer corners and edges that could be disturbing and contributes to the smooth and robust overall design of the mobile cooling box according to the present embodiments.

(18) According to another embodiment of the mobile cooling box, the hinge pin has a tool

engagement portion at its front end for fastening the pin module. This further eases strong fixation of the pin module. The kind of tool engagement portion is not particularly limited. It can be, for example, an outer contour at the hinge pin's front end in the form of a hexagon screw head. However, other forms of tool engagement portions are also possible within the scope of the present embodiments.

(19) According to a further embodiment of the mobile cooling box, the tool engagement portion is a hexagonal socket at the front end face of the hinge pin and the entire hinge pin has a smooth outer surface of a cylindrical shape. Apart from being optically more esthetic, since the entire outer surface of the hinge pin is a smooth cylinder, the hinge construction and the respective pivotal guiding of the lid is more resilient.

(20) According to one embodiment of the mobile cooling box, the entire pin module including the hinge pin, the backing plate and the engaging portion may be formed of metal as one single and integral component. This makes the pin module more robust and contributes to the smooth overall design of the mobile cooling box.

(21) According to another embodiment of the mobile cooling box, the bearing module is present at the lid and the pin module is present at the box main body. Furthermore, the hinge bearing only partly envelops the hinge pin and is open in a direction perpendicular to the longitudinal axis of the hinge pin. Thus, the bearing module allows the hinge pin to be released from the hinge bearing, thereby enabling the lid to be removed completely from the box main body. Removing the lid is of advantage in a number of situations, for example, for cleaning purposes. With one side open of the otherwise pin-surrounding or pin-enveloping hinge bearing, the lid can simply be pulled away from the pivot axis, without the need of any further actions, possibly with only one hand.

(22) According to another embodiment of the mobile cooling box, the bearing module may be configured so that the lid cannot be removed from the box main body when the mobile cooling box is closed. Thus, removing the lid is only possible when the box is open which prevents unintentional lid removal and warming of the inside of the mobile cooling box.

(23) According to a further embodiment of the mobile cooling box, the bearing module may further comprise a spring element configured to hold the hinge pin in place and to provide a predefined and limited resistance when removing the lid from the box main body. Thus, a slight additional force has to be exerted for removing the lid against said resistance. This further prevents the lid from unintentionally falling off the box main body, when the box is open.

(24) According to another embodiment of the mobile cooling box, the bearing module further comprises an abutting portion. The abutting portion, when the lid is pivoted open for a predetermined angle of more than 90° , abuts against a region of the box main body, thereby enabling the lid to rest in the open position. Thus, the lid can stand in a more or less upright position without being held by the user, who, thus, has his or her hands free for loading or unloading the mobile cooling box. Generally, in this embodiment the bearing module with its abutting portion is configured so that the lid can stay in an open position with an angular deflection from the closed position in the range of 91° to 180° , more specifically 91° to 150° and even more specifically 91° to 120° .

(25) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. All of the above outlined features are to be understood as exemplary only and many more features and objectives of the various embodiments may be gleaned from the disclosure herein. Therefore, no limiting interpretation of this summary is to be understood without further reading of the entire specification, claims and drawings, included herewith. A more extensive presentation of features, details, utilities, and advantages of the present invention is provided in the following written description of various embodiments of the invention, illustrated in the accompanying drawings, and defined in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the following, embodiments of the mobile cooling box are described in more detail with reference to the accompanying drawings, wherein
- (2) FIG. 1 shows a front perspective view of a mobile cooling box according to present embodiments;
- (3) FIG. 2 shows a back-perspective view of the mobile cooling box of FIG. 1;
- (4) FIG. 3 shows the open mobile cooling box of FIG. 1;
- (5) FIG. 4 shows a front perspective view of another mobile cooling box according to present embodiments;
- (6) FIG. 5 shows the open mobile cooling box of FIG. 4;
- (7) FIG. 6 shows a front perspective view of another mobile cooling box according to present embodiments;
- (8) FIG. 7 shows the open mobile cooling box of FIG. 6;
- (9) FIG. 8 shows an isolated perspective view of a user interface module;
- (10) FIG. 9 shows an exploded view of the component of FIG. 8;
- (11) FIG. 10 shows another exploded view of the component of FIG. 8;
- (12) FIGS. 11 to 13 illustrate a sequence of a mounting procedure;
- (13) FIG. 14 shows a section of the mobile cooling box of FIG. 1 with actuated latch handle;
- (14) FIG. 15 shows an exploded view of the latch handle of FIG. 14;
- (15) FIG. 16 shows an isolated perspective view of the latch handle of FIG. 14;
- (16) FIG. 17 illustrates the working principle of the latch handle of FIG. 14;
- (17) FIG. 18 illustrates the mounting procedure of a handle module;
- (18) FIG. 19 shows an exploded view of the handle module of FIG. 18;
- (19) FIGS. 20 and 21 show different perspectives of a cut view of the outer side wall;
- (20) FIG. 22 shows a section of the mobile cooling box of FIG. 1 with a hinge module;
- (21) FIG. 23 shows relevant parts of FIG. 22;
- (22) FIG. 24 shows an inside perspective view of the hinge module of FIG. 22;
- (23) FIG. 25 shows a section of the mobile cooling box of FIG. 1 with removed lid;
- (24) FIG. 26 shows the section of FIG. 25 and illustrates a mounting procedure;
- (25) FIG. 27 illustrates the insertion of an ice maker module into the open mobile cooling box of FIG. 4;
- (26) FIGS. 28 and 29 show different perspectives of an exploded view of the ice maker module; and
- (27) FIG. 30 shows a section of the open mobile cooling box of FIG. 4 with a lamp system.

DETAILED DESCRIPTION

- (28) The illustrated mobile cooling boxes 1 in FIGS. 1 to 7 are generally rectangular in shape. Basically, the mobile cooling boxes according to some embodiments have a box main body 2 and one or a plurality of lids, for example two lids 3 for opening the box 1 and providing access to the inside of the box 1. In the present case, access to the inside of the box 1 is possible from above, but is not limited thereto. The front edge of the lid 3 can be pivotally opened. The rear edge is hinged to the box main body 2. At its front and rear edges, the mobile cooling box 1 is rounded, while the side edges are covered and protected by a fender frame 23 that forms part of the box main body 2. The height of the fender frame 23 is equal to the level of the lid 3 when the mobile cooling box 1 is closed. Thus, the lid 3 when being closed sort of sinks or recesses between the two opposite fender frames 23 thus offering a smooth, uniform and robust look of the mobile cooling box.
- (29) In this context and within the framework of the present embodiments, but without limitation, all directional terms, like front, rear, back, upper, lower, above, sink, as well as broadness and depth

refer to the mobile cooling box **1** standing on the ground as usually intended and from a perspective facing the side of the mobile cooling box **1** where the edge of the lid is pivotable to the above while opening, unless explicitly stated otherwise.

(30) Each illustrated mobile cooling box **1** is of different depth and width. The lid **3** or the lids **3** are to be opened from a side where the fender frame **23** is not located. This is in case of the mobile cooling box **1** of rather small size, as illustrated in FIGS. **1** to **3**, the shorter side of the mobile cooling box **1**. In case of the two mobile cooling boxes **1** of rather large size, as illustrated in FIGS. **4** and **5** and FIGS. **6** and **7**, it is the longer side of the mobile cooling box **1**, respectively.

(31) The mobile cooling box **1** has an electrically driven cooling unit and comprises an internal battery (not shown). The mobile cooling box **1** can be used in plugged-in mode or in battery mode. The mobile cooling box **1**, therefore, has the required sockets **24** located at one of its sides.

(32) In the following, different aspects and features of the mobile cooling box are described. As will become apparent, many of the following aspects relate to readily mountable modules for various functions which can be mounted without limitation to the mobile cooling box **1** regardless of the size thereof.

(33) The FIGS. **8** to **10** show a user interface. With such a user interface it is possible for the user to sort of communicate with the mobile cooling box **1**, that is, retrieving information from the mobile cooling box **1** and entering controls into the mobile cooling box **1**. The mobile cooling box **1** is equipped with the user interface module **100** for operation and control by the user. The user interface module **100** is mounted at the mobile cooling box **1** in a way that a part of it is engaged with a designated opening at the mobile cooling box **1** on one side of the user interface module **100** and fixed with additional fixation structure at the other side of the user interface module **100**. For example, as illustrated, screws **170** are used for fixing the module **100** at the left side, however other structures may be utilized. On the right side, no screw is necessary. At this side the module **100** is engaged with the designated opening at the mobile cooling box.

(34) As shown in FIGS. **9** and **10**, the user interface module **100** is an assembly of components, namely a circuit board **110**, a housing **120** and a front cover **130**. The assembly is mounted in a recessed part of the mobile cooling box **1** so that essentially only the front cover **130** is directly visible for the user.

(35) The circuit board **110** forms a latch **111** extending away from the right side of the assembly with respect to the housing **120** and the front cover **130**. The latch **111** is engaged with the designated opening at the mobile cooling box **1**. The circuit board **110** is essentially longer in size than the housing **120** at the right side. The circuit board **110** extends over the edge of the housing **120** and the front cover **130**. In the illustrated embodiment, the circuit board **110** extends over the edge of the housing **120** and the front cover **130** for about 1 cm, but also other dimensions are possible. At the backside of the circuit board **110** the part forming the latch **111** is further strengthened by additional material provided in this area.

(36) As shown in FIG. **10**, the circuit board **110** has a connector **112**. The connector **112** is located at the backside of the circuit board **110**. The user interface module **100** is electronically connected with the mobile cooling box **1** by using the connector **112**. A respective plug (not shown) is provided in the inside of the box main body **2** of the mobile cooling box **1** and can be reached from the opening in which the user interface module **100** is hooked.

(37) The user interface module **100** further has a USB port **113**. In the illustrated embodiment a single USB port **113** is provided. However, there can be also a plurality of USB ports provided, for example depending on the size of the mobile cooling box. The USB port **113** is present at the circuit board **110**. Moreover, two through holes for a screw connection are provided. By using the USB port **113** the user can recharge external devices like batteries, lamps, smartphones, etc. Moreover, the USB port **113** provides access to the internal control of the mobile cooling box **1** and, depending on the settings and version, enables download of internal data and/or programming of functions of the mobile cooling box **1**.

(38) The housing **120** is clamped onto or over the circuit board **110** by respective clip-in elements **121**. In the shown example, three clip-in elements **121** on each of the upper and the lower side of the housing **120** are sufficient to achieve a stable and robust fixation of the housing **120**. The housing **120** is formed of an injection molded plastic component. The housing provides co-injected regions of plastic that is softer than at other regions of the housing **120**.

(39) At the housing **120**, the area of the USB port **113** is left open to allow access to the USB port **113**. A rubber cap **140** is provided for covering the USB port **113** when not in use. In the present embodiment, the rubber cap **140** is swingably attached to the housing **120** to avoid losing the rubber cap **140**. By removing the rubber cap **140** from the housing **120** access to the USB port **113** becomes possible. Here, the size and design of the rubber cap **140** ensures coverage of the through holes provided for the screw connection at the same time.

(40) The user interface module **100** further has a display **160**, the display **160** is arranged behind the front cover **130** and the front cover **130** of the user interface module **100** is transparent at least in the area of the display **160**.

(41) The user interface module **100** further has three operation devices, here in the form of buttons **150** extending from the user interface module **100**. One of the buttons **150** is located at the right side of the display **160** and provides an up-and-down selection button **150** for navigating through the menu of the control menu of the implemented software. In the illustrated embodiment, two further buttons **150**, here designed in form of single round buttons, are provided at both sides next to the display **160**, or, respectively, the afore-mentioned button **150**. The buttons **150** are made of rubber, or covered by rubber, in order to provide good haptics and provide a stable and robust design.

(42) The front cover **130** user interface module **100** is of a scratch resistant material or has a scratch resistant coating.

(43) The user interface module **100** is capable of wireless communication with an external electronic device, for example by Bluetooth technology, WLAN or any other suitable technology. The external electronic device can be a remote control, a smartphone or the like. Hence, the user interface **100** and thus the mobile cooling box is remote controllable with the external electronic device. In case of using a smartphone, a respective app is available and to be used on the smartphone.

(44) The user interface module **100** provides to the user functions and controls like ON/OFF-switching the cooling, temperature control including setting, if desired in a time-shift manner, displaying current temperature, temperature history graphs, temperature type setting ($^{\circ}$ C./ $^{\circ}$ F.), alarm setting, energy saving mode, displaying battery status, including voltage level and/or battery remaining time, power consumption history graphs, lid **3** open indication, wireless communication ON/OFF and setting display brightness. The below list shall not be construed as conclusive. Further functions are, thus, also possible.

(45) For mounting the user interface module **100**, the part on the right side of the user interface module **100** that is supposed to be engaged with the designated opening at the mobile cooling box **1** is laterally slid into the designated opening. This step is illustrated in FIG. **11**. In FIG. **11** the rubber cap **140** is shown in its open position. However, the rubber cap **140** itself is not involved in the inserting and assembling process of the user interface module **100**. After being inserted with the latch **111**, the left side of the user interface module **100** is rotated in place as shown in FIG. **12** and, in the next step, the user interface module **100** is fixed with two screws to complete the mounting process.

(46) The mobile cooling box **1** according to some embodiments and as described before has at least one lid **3**. By using the lid **3** the mobile cooling box **1** can be opened from one side-edge of the lid **3**. Thereby, access is provided to the inside of the box **1**. At the opposite side-edge of the lid **3**, the lid **3** is hinged to the box main body **2**. By this hinge connection the lid **3** can be pivoted upwards.

(47) As shown in FIGS. **14** to **17**, the mobile cooling box **11** is equipped with a latch handle module

200. The latch handle module **200** allows manually locking and unlocking of the lid **3** and, thus, opening and closing the mobile cooling box **1** by the lid **3**. The latch handle module **200** is integrated in the lid **3** and located at the front side edge of the lid **3** of the illustrated embodiment of the mobile cooling box **1**.

(48) The latch handle module **200** is an assembly of components including an actuating element **201**, a locking element **202** and a casing **203**. The actuating element **201** is manually operable by the user. The locking element **202** is engageable with a corresponding counterpart at the box main body **2**. By engaging the corresponding counterpart at the box main body **2** the lid **3** is locked from being opened.

(49) The latch handle module **200** provides a mechanism for locking and unlocking the lid **3**. According to the latch handle module **200** the actuating element **201** and the locking element **202** are mechanically connected to each other. As illustrated in FIG. **17**, by operating the actuating element **201**, the locking element **202** disengages with its corresponding counterpart at the box main body **2** and allows up-folding the lid **3**. In this embodiment, the locking element **202** is designed as a snap-in latch. The snap-in latch, in a closed state, extends into the corresponding counterpart being a recess in the box main body **2**. Furthermore, the actuating element **201** has a surface which can be pushed by the fingers of the user's hand. In the illustrated embodiment the actuating element **201** has a width of about 10 cm. However, according to the overall size the width of the actuating element **201** can have also a different size. For opening the lid **3**, the actuating element **201** is pivoted about an axis of rotation with a pivoting direction that is the same as that of the lid **3** when being opened. Therefore, there are no opposing movements for the user's hand, which has been found to be comfortable for the user.

(50) As regards the working principle of the latch handle module **200**, the latch handle module **200** further comprises a shaft **204**. The shaft **204** has a longitudinal axis being co-linear with the axis of rotation of the actuating element **201**. The actuating element **201** is connected to and pivotable about the shaft **204**. The shaft **204** is of a rigid metal material and extends essentially over the entire width of the latch handle module **200**. The latch handle module **200** further has two springs **205** by means of which the mechanism provided by the latch handle module **200** is spring loaded. The mechanism provided by the latch handle module **200** is spring loaded for providing a restoring force that ensures that the actuating element **201** and the locking element **202** return to their respective initial positions after an operation of the actuating element **201** by the user.

(51) As shown in FIGS. **18** and **19**, the mobile cooling box **1** is equipped with two handle modules **300**. The two handle modules **300** are located at an outer side surface of the box main body **2**. One handle module **300** has a handlebar **301**. The handlebar **301** is intended to be grasped by the hand of the user and has a longitudinal axis as well as two ends, two hangers **302** and two brackets **303**. The handlebar **301** is attached at its two ends to the two hangers **302**. The hangers **302** are rotatably mounted at the two brackets **303**. The two brackets are fixed to the outer side surface of the box main body **2**.

(52) The handle module **300** is designed in a way that the handlebar **301** hangs downwards in an unactuated state and can be swung out and upwards for carrying the mobile cooling box **1**.

(53) Each of the brackets **303** comprises a mounting area, or mount, **304** and a shielding area, or shield, **305**. The mounting area **304** faces the outer side surface of the box main body **2** to which the bracket **303** is fixed. The shielding area **305** hides the hangers **302** and the handlebar **301** in an unactuated state of the handle module **300** and in a lateral perspective along the longitudinal axis of the handlebar **301**.

(54) The handle module **300** is designed so that, in an unactuated state of the handle module **300** and in a lateral perspective along the longitudinal axis of the handlebar **301**, at least a section of the outer contour of the shielding area **305** is flush with the handlebar **301** and with the hangers **302**. Thus, when the mobile cooling box **1** is not carried, the handlebar **301** with its hangers **302** exactly hides behind the bracket **303** in the respective lateral perspective.

(55) The handlebar **301** and its hangers **302** are spring-loaded. Thus, in an unactuated state, the handlebar **301** and the hangers **302** are forced in a direction to the mobile cooling box **1** and are thus kept hidden in-between the shielding areas **305** of both brackets **303**. For this purpose, two springs **308** are arranged within the handle module **300**. The springs **308** force the hangers **302** relative to the brackets **303** to abut against the part with the mounting area **304**.

(56) The handle module **300** is designed in a way that, in an actuated state, the hangers **302** with the handlebar **301** are swung out and upwards and rest in a position relative to the mobile cooling box **1**. Thus, the mobile cooling box **1** can be carried in a comfortable way. The hangers **302** with the handlebar **301** rest in the position by means of a region of the hangers **302** abutting against a region of the brackets **303**. Thereby, at the joint between the brackets **303** and the hangers **302**, the hangers are rounded in a section around the respective pivot axis. Moreover, a corresponding roundness is present at the brackets **303** to the extent that, when the hangers pivot out, the round part of the brackets **303** that enclose the round part of the hangers abut against the flanks of the hangers **302**. Thus, further rotation of the hangers **302** is blocked.

(57) Furthermore, at its mounting area **304** each bracket **303** comprises two through holes **306** for fixing the bracket **303** to the outer side surface of the box main body **2** by means of fixing elements **307**. In the illustrated embodiment of the mobile cooling box the fixing elements are designed in the form of screws but are not limited thereto. The through holes **306** and the respective fixing elements **307** are covered by the hanger **302** that is mounted to said bracket **303**, in an unactuated state of the handle module **300**. Thereby, the hanger **302** abuts against said mounting area **304**.

(58) An additional accessory, like for example a bottle opener (not shown) or other equipment or tooling, can be attached at the through holes **306** by respective means, like for example screws **309**.

(59) As mentioned, the mobile cooling box **1** is basically rectangular in shape and has different dimensions in width and depth and height. Further, the two handle modules **300** are located at the respective two shorter outer side surfaces of the mobile cooling box **1** being opposite to each other. Thereby, when carrying the mobile cooling box **1** a tilting of the mobile cooling box **1** can be avoided.

(60) In the present embodiment the handlebar **301** has a circular cross-section. Moreover, the handlebar **301** has a length of at least 10 cm to ease gripping the handlebar by the user's hand. However, other dimensions are also possible. The lower part of the hangers **302** correspond with this rounded contour. Also, the lower part of the brackets **303** partly correspond with this contour. Hence, the components are flush in an unactuated state.

(61) At least the handlebar **301**, the hangers **302** and the brackets **303** of the handle module **300** are made of aluminum. At least part of the surface of the aluminum is roughened and has an oxidic protective layer.

(62) As illustrated in FIGS. **20** and **21**, the mobile cooling box **1** is equipped with air vents **400**. The air vents **400** are located on at least one side wall of the box main body **2**. In the periphery of the air vents **400** cord fixation means are present (not shown). The electrical cord (not shown) provided for connecting the mobile cooling box to electrical power can be, especially in case the cord is not in use, attached to the outside of the mobile cooling box **1** in a known manner. The cord fixation means can for example be formed in the shape of hooks to which the cord can be removably attached. Nearby the air vents **400**, as illustrated in FIG. **20**, at least one power connector is present to connect the removable power cord (not shown) to the mobile cooling box **1** to supply electrical power to the mobile cooling box **1**.

(63) The air vents **400** comprise a plurality of horizontal opening or slots **401** (in the following generally referred to as slots), respectively, allowing air circulation through the respective side wall of the mobile cooling box **1**. The slots **401** comprise shielding elements **402** protruding inside the mobile cooling box **1**. Each of the shielding elements **402** is designed in such a way that the shielding element at least partly blocks the view into the inside of the mobile cooling box **1** from the outside. In other words, the inside of the mobile cooling box **1** is not visible from the outside

due to the design of the shielding elements **402**.

(64) One slot **401** has an upper edge **403** and a lower edge **404**. Both, the upper edge **403** and the lower edge **404** lie in the plane of the respective side wall. One of the shielding elements **402** extends from the lower edge **404** to the inside of the mobile cooling box **1** and further upwards with respect to said lower edge **404**, virtually in the direction of and at least up to the height of the upper edge **403**. Thus, the inside of the mobile cooling box **1** is not visible from the outside due to the design of the shielding element **402**

(65) Particularly, in a vertical cross section perpendicular to said side wall, the shielding element **402** extends from the lower edge **404** in upward curved form, namely in the form of a segment of a circle.

(66) Furthermore, one of the shielding elements **402** extends from the upper edge **403** to the inside of the mobile cooling box **1**.

(67) Particularly, in a vertical cross section perpendicular to said side wall, the shielding element **402** extends from the upper edge **403** to the inside of the mobile cooling box **1** in a straight horizontal direction. This has essentially the function of providing more stability to the side wall and to uniform the upper and lower edges **403** and **404** with regard to the roundness.

(68) The side wall where the vents are present is manufactured together with the shielding elements **402** as a one-piece component which is made of plastic and manufactured by injection molding.

(69) As is shown in FIGS. **22-26**, the mobile cooling box **1** has a lid **3**. The lid **3** is pivotally attached to the box main body **2** by means of two hinge modules **500**. Each hinge module **500** comprises a pin module **510**. The pin module **510** has a hinge pin **511** with a front end, a rear end, a longitudinal axis about which the lid **3** is pivotable, and a smooth outer surface having a cylindrical shape.

(70) The hinge module **500** further comprises a bearing module **530**. The bearing module **530** has a hinge bearing **531** accommodating the hinge pin **511**. The hinge pin **511** laterally extends with its front end into the hinge bearing **531**. Thus, during pivoting the lid **3** with respect to the box main body **2** an axis of the hinge bearing **531** remains co-linear with the longitudinal axis of the hinge pin **511**.

(71) For the mounting of the pin module **510**, the pin module **510** further comprises an engaging portion **513**, here in form of a bolt portion **513**. The bolt portion **513** has a male thread and extends from the rear end of the hinge pin **511**. The bolt portion **513** has a longitudinal axis being co-linear to that of the hinge pin **511**.

(72) The pin module **510** further comprises a backing plate **514** between the hinge pin **511** and the bolt portion **513**. The backing plate **514** lies in a plane perpendicular to the longitudinal axis of the hinge pin **511** and has a pin-side surface and a bolt-side surface. The backing plate **514** has a circular shape so that it is symmetrical with regard to rotation.

(73) The pin module **510** is mounted to the box main body **2** at a vertical surface thereof which is the inner side of a part of the box main body **2**. The backing plate **514** abuts with its bolt-side surface against said vertical surface of the box main body **2**.

(74) Furthermore, the vertical surface of the box main body **2** to which the pin module **510** is attached to has a pin module attachment portion **520**. The pin module attachment portion **520** comprises a bore **521** having a female thread, in which the bolt portion **513** is fastened, and a recess **522** for accommodating the backing plate **514**. The recess **522** has a depth corresponding to the thickness of the backing plate **514**. Hence, the transition from said vertical surface of the box main body **2** to the surface of the pin-side surface of the backing plate **514** is flush. In order to provide for sufficient stability, the thickness of the backing plate **514** is about 2 mm.

(75) Furthermore, the hinge pin **511** has a tool engagement portion **512** at its front end for fastening the pin module **510**. The tool engagement portion **512** is a hexagonal socket that is engageable with a hex key at the front end face of the hinge pin **511**. Moreover, the entire hinge pin **511** has a smooth outer surface of a cylindrical shape, so that the pivoting movement can be guided over the

entire length of the hinge pin **511**.

(76) The entire pin module **510** including the hinge pin **511**, the backing plate **514** and the bolt portion **513** is formed of metal. Moreover, the entire pin module **510** is formed as one single and integral component. Thus, the pin module **510** is very robust component.

(77) The bearing module **530** is present at the lid **3** and the pin module **510** is present at the box main body **2**. The hinge bearing **531** only partly envelops the hinge pin **511** and is open in a direction perpendicular to the longitudinal axis of the hinge pin **511**. Thus, the bearing module **530** allows the hinge pin **511** to be released from the hinge bearing **531**, thereby enabling the lid **3** to be removed completely from the box main body **2**. In particular, when it is pivoted in an open direction for about 60° and more the lid **3** can be removed. Thus, the bearing module **530** is configured so that the lid **3** cannot be removed from the box main body **2** when the mobile cooling box **1** is closed.

(78) The bearing module **530** further comprises a spring element **533**. The spring element **533** protrudes out of an upper surface part of the hinge bearing **531**. The spring element **533** is configured to hold the hinge pin **511** within the hinge bearing **531** and to provide a certain resistance during removing the lid **3** from the box main body **2**.

(79) The bearing module **530** further comprises an abutting portion **532**. When the lid **3** is pivoted in the open direction for an angle of about 100° the abutting portion **532** abuts against a region of the box main body **2**. Thereby, the lid **3** is enabled to rest in an open position.

(80) As shown in FIGS. **27** to **29** the mobile cooling box **1** has an inside that is laminated with a lining **601** at the inner side walls **21** and at the floor **22** of the box main body **2**.

(81) The mobile cooling box **1** is equipped with an ice maker module **600**. The ice maker module **600** has a freezing compartment **606**. The ice maker module **600** can be removably placed on a freezing zone **602** on a floor part of the lining **601**.

(82) The mobile cooling box **1** further comprises an evaporator **603** arranged underneath the lining **601** at the freezing zone **602**, for providing sufficient cooling power for freezing goods.

(83) The ice maker module **600** is an assembly of components, namely a frame **604** and a cover **607**. The frame **604** has lateral walls **605** limiting the freezing compartment **606**. The cover **607** is attached to the upper side of the frame **604** for opening and closing the ice maker module **600** and providing access from above to the freezing compartment **606**. The freezing compartment **606** is limited at its ground by the lining **601** at the freezing zone **602**. Thus, the goods to freeze are placed directly on the floor part of the freezing zone **602** for efficient freezing.

(84) The freezing zone **602** is rectangular and is located in a niche limited by the lining **601** of three of the inner side walls **21**. The ice maker module **600** fits in the niche.

(85) At least one pair of corresponding attachment means **608** configured to releasably engage with each other is present at the lining **601** of the inner side walls **21** adjacent to the freezing zone **602** and at the ice maker module **600**, respectively. By the at least one pair of corresponding attachment means **608** the position of the ice maker module **600** is secured. The pair of attachment means **608** provides for a form-locked connection being a snap-in connection. The snap-in connection consists of hook and a corresponding recess. The hook is a projecting element that is configured to snap in the recess. The hook is located at the ice maker module **600** and the corresponding recess is located at the lining **601** of the respective inner side wall **21**. The hook is located at the frame **604** of the ice maker module **600**.

(86) The hook and the recess of one pair of corresponding attachment means **608** are formed as integral parts of the lining **601** and the ice maker module **600**, respectively.

(87) Furthermore, the cover **607** is hinged to the frame **604**. Thus, the cover **607** is swingably openable to the above and can be opened about an angle of about 100°. The cover **607** has a grip portion **609** by means of which the cover **607** can be opened and closed by the hand of the user.

(88) The ice maker module **600** further comprises two ice trays **610**. The ice trays **610** fit into the freezing compartment **606**. Each of the ice trays **610** is equipped with a cap **611**. Each ice tray **610**

has a plurality of recesses for forming ice cubes. The cap **611** has small holes **612** in form of bores with a rather small diameter. By these holes air exchange is enabled between inside and outside of the ice tray, but predominantly preventing water from leaking out.

(89) The opening of the tiny holes has a cross section of about 0.20 mm. Above each recess, one of the tiny holes is arranged.

(90) As shown in FIG. **30** the mobile cooling box **1** comprises a lamp module **700**. By the lamp module **700** light can be provided in the inside of the box main body **2**. To turn on the lamp module **700** it does not have any mechanical switches as in usual refrigerators. In the present embodiment of the lamp module **700** can be switched ON or OFF by means of a reed sensor (not shown). The front cover of the lamp module **700** is perfectly flush with the surface at which the lamp module **700** is arranged.

(91) The front cover is mounted to the inner lining in a waterproof manner. Specifically, the front cover of the lamp module **700** is clipped in a corresponding recessed part of the inner lining and is equipped with sealed portions.

(92) Furthermore, the front cover of the lamp module **700** is transparent and provides a diffuse light. The light is emitted from diodes inside the lamp module **700** and both, the light-emitting diodes and the reed sensor are mounted on a circuit board of the lamp module **700**.

(93) For switching the light ON and OFF, a magnet is incorporated in the part of the lid **3** that functionally corresponds with the reed sensor. In the closed state of the lid **3**, the magnet is located in the vicinity of the light module **700** so that the light module is switch OFF. While opening or in the opened state the distance of the magnet, thus, is increased and the light module is switch ON by the reed sensor.

REFERENCE SIGNS

(94) **1** Mobile cooling box **2** Box main body **3** Lid **21** Inner side wall of box main body **22** Floor of box main body **23** Fender frame **24** Socket **25** Outer side wall of box main body **100** User interface **110** Circuit board **111** Latch **112** Connector **113** USB port **120** Housing **121** Clip-in element **130** Front cover **140** Rubber cap **150** Operation device/button **160** Display **170** Screw **200** Latch handle module **201** Actuating element **202** Locking element **203** Casing **204** Shaft **205** Spring at the latch handle **300** Handle module **301** Handlebar **302** Hanger **303** Bracket **304** Mounting area **305** Shielding area **306** Through hole **307** Fixing elements **308** Springs at the handle **309** Screws of the handle **400** Air vents **401** Opening/slot **402** Shielding element **403** Upper edge **404** Lower edge **500** Hinge module **510** Pin module **511** Hinge pin **512** Tool engagement portion **513** Engaging portion/bolt portion **514** Backing plate **520** Pin module attachment portion **521** Bore **522** Recess **530** Bearing module **531** Hinge bearing **532** Abutting portion **533** Spring element **600** Ice maker module **601** Lining **602** Freezing zone **603** Evaporator for the freezing zone **604** Frame **605** Walls of frame **606** Freezing compartment **607** Cover of ice maker module **608** Attachment means **609** grip portion **610** Ice tray **611** Ice tray cap **612** Tiny holes **700** Lamp module

Claims

1. A mobile cooling box, comprising: a box main body, having a first pair of parallel walls and a second pair of parallel walls extending from a bottom of said box main body; upper ends of said walls defining an opening wherein a lid is positioned, said lid pivotable between an opened position and a closed position to access an interior of the box main body; a first hinge module and a second hinge module disposed between said lid and said box main body; each of said first hinge module and said second hinge module having a hinge pin which is fixed relative to said box main body and engages said lid when said lid is in said closed position; a backing plate, said hinge pin extending from said backing plate; an engaging portion extending opposite from said hinge pin, said engaging portion extending into a wall of said box main body; each said backing plate positioned against one wall of said walls; a hinge bearing which engages said hinge pin, said hinge

bearing partly enveloping said hinge pin so that said hinge pin may be selectively released from said hinge bearing.

2. The mobile cooling box of claim 1, said engaging portion comprising a thread.

3. The mobile cooling box of claim 1, said engaging portion defined by a bolt.

4. The mobile cooling box of claim 1, said engaging portion being co-linear to said hinge pin.

5. The mobile cooling box of claim 1 the hinge pin having a smooth outer surface.

6. The mobile cooling box of claim 1, wherein said hinge bearing is capable of being selectively released from said hinge pin when said lid is opened a preselected amount or to a preselected position.

7. The mobile cooling box of claim 1, said first pair of parallel walls being longer than said second pair of parallel walls.

8. The mobile cooling box of claim 1, said lid pivoting along one wall of said longer walls or one wall of said shorter walls.

9. The mobile cooling box of claim 1, said hinge bearing having an opening at which said hinge pin may be released.

10. The mobile cooling box of claim 9, further comprising a retention structure extending into said opening.

11. The mobile cooling box of claim 10, said retention structure being biased to one position.

12. The mobile cooling box of claim 11, said retention structure being movable.

13. The mobile cooling box of claim 12, said retention structure being a spring.

14. The mobile cooling box of claim 1, said lid being a first lid and a second lid.

15. The mobile cooling box of claim 1, said walls being two long walls and two short walls.

16. The mobile cooling box of claim 14 further comprising a floor having a first elevation and a second elevation.

17. The mobile cooling box of claim 16, said second elevation being higher than said first floor elevation, wherein cooling components are disposed beneath said floor at said second elevation.

18. The mobile cooling box of claim 17, further comprising a lining disposed along an interior of said box main body.
